

****Continuum Equations for Heterogeneous Reaction with transport**

*Heterogeneous reactions are the chemical reactions taking place on a solid surface, referred to as a catalyst.

*Heterogeneous reactions are also called catalytic reactions.

*Catalyst is a substance (solid/liquid/gas) that reduces activation barrier of a reaction

*Catalytic reaction can be either homogeneous or heterogeneous reactions

*Examples of homogeneous catalytic reactions:

1. Fischer-Esterification



Carboxylic acid Alcohol Esters

Esters are used to produce soap, perfumes and food products.

2. Friedel-Crafts Acylation



Arene

Ketone

Ketone is most used as solvent in various processes.

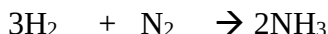
3. Monsanto Process: Catalytic carbonylation of methanol



Use: solvent, dyes, perfumes, plastic, polymer, medical, food.

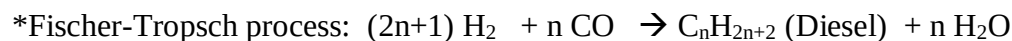
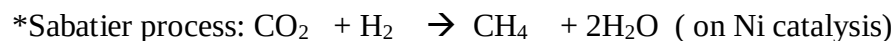
*Examples of Heterogeneous Catalytic Reactions

1. Haber-Bosch Process (gas-solid)



Use: fertilizers

2. Catalytic hydrogenation reaction



3. Catalytic cracking

Used to convert high-boiling hydrocarbons fractions of petroleum crude into gasoline or more valuable olefinic gases.

Long chains Alkanes $\rightarrow$ smaller alkanes

$\rightarrow$ Cyclo alkanes

$\rightarrow$ Branched alkanes

$\rightarrow$ Aromatics

$\rightarrow$ Smaller or branched alkanes

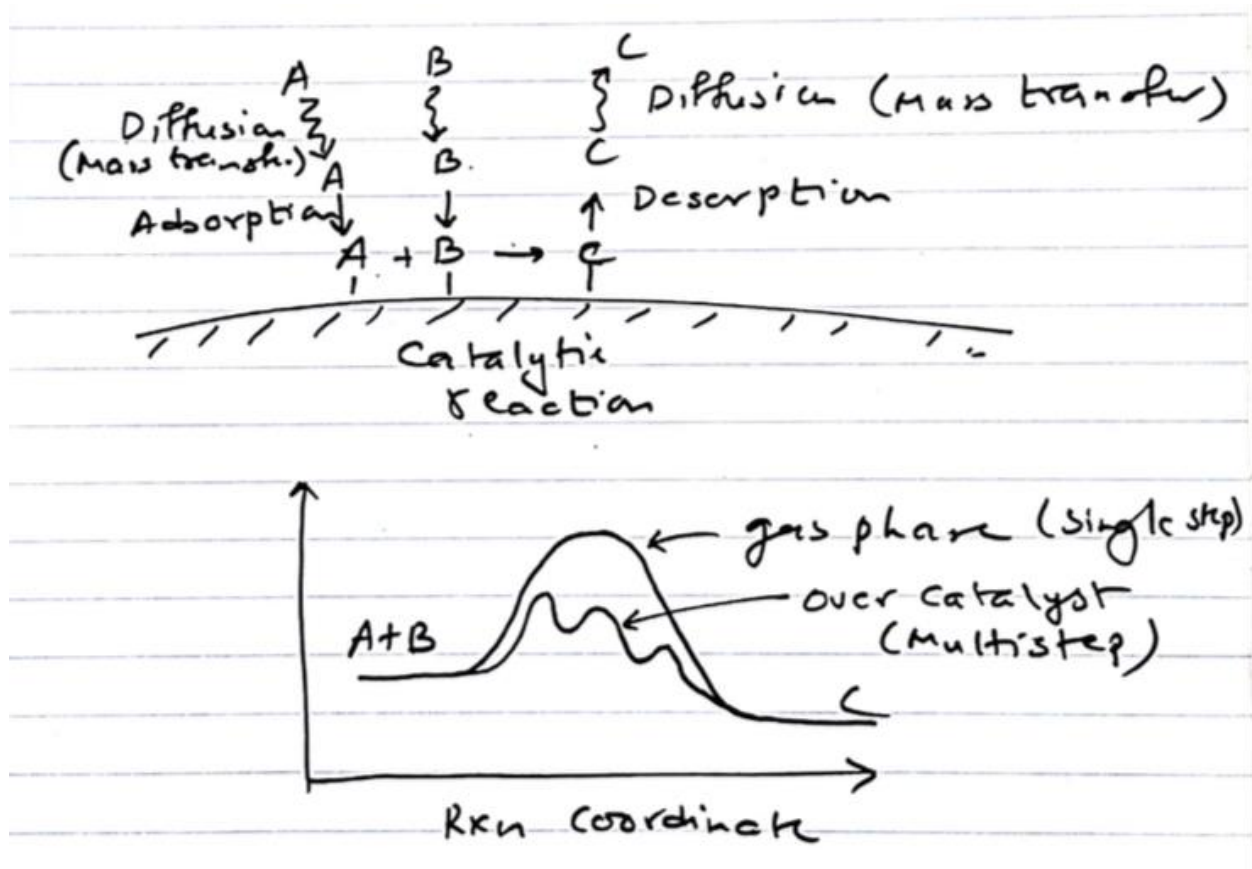
Zeolite-Aluminosilicate.

4. Photocatalytic or Electrocatalytic

Waste water + H_2O_2 (O_2) $\rightarrow$ clean water.

****Physical processes involved in heterogeneous catalytic reaction**

*Consider a reaction $\text{A} + \text{B} \rightarrow \text{C}$ over a catalyst surface.



*Continuum equations for homogeneous catalytic reactions are similar to homogeneous non-catalytic reactions.

*In case of heterogeneous catalytic reactions, there could be two different cases.

1. Catalyst is dispersed throughout the bulk reaction mixture.

For a general reaction

$$\sum_{j=1}^s \alpha_{ij} A_j = 0, \quad i = 1, 2, \dots, m$$

Reaction rate is defined per unit area

$$\hat{r} \left(\frac{\text{moles}}{\text{area} \times \text{time}} \right) = \sum_{i=1}^m \alpha_{ij} \hat{r}_i \quad (46)$$

The continuity equation can be written as

$$\frac{DC_j}{Dt} + \underline{\nabla} \cdot \underline{J_j} = a_v \sum_{i=1}^m \alpha_{ij} \hat{r}_i - C_j \frac{D \ln \hat{V}}{Dt} \quad (47)$$

Where a_v is the concentration of active surface area. It can be viewed as (concentration of active sites) x (area of sites). $a_v \equiv \frac{Area}{Volume}$

*When active sites are reaction-specifics, the reaction term in (47) is $\sum_{i=1}^m \alpha_{ij} \hat{r}_i$

Ex: Fluidized/packed bed reactor.

2. Catalyst is present only at the boundary of the reaction domain.

Ex: stirred tank or tubular reactor with catalyst on the wall. Electrocatalytic reaction.

The continuity equation becomes:

$$\frac{DC_j}{Dt} + \underline{\nabla} \cdot \underline{J_j} = - C_j \frac{D \ln \hat{V}}{Dt} \quad (48)$$

With boundary condition

$$\underline{n} \cdot \underline{N_j} = \sum_{i=1}^m \alpha_{ij} \hat{r}_i \quad (49)$$

****Heterogeneous Reaction with consumption of solid surface**

*There are many heterogeneous reactions that results in consumption or deactivation of solid surface.

*Ex: 1. Combustion reaction $\rightarrow$ coking of surface

2. Precipitation reaction $\rightarrow$ solid deposition.

*In such a case, the boundary condition, $\underline{n} \cdot \underline{N_j} = \sum_{i=1}^m \alpha_{ij} \hat{r}_i$ must be accompanied by an equation of the moving surface.

*Suppose we describe the surface at any time t by $f(x,t)=0$



*Say $\underline{x}$ on surface at time t move to a new point as a result of the surface receding to a new location.

Let $\alpha \equiv$ Volume of catalyst consumed per unit moles of species reacted.

$\alpha \hat{r} \equiv$ Length travelled by point $\underline{x}$ per unit time.

$\underline{n} = \frac{\nabla f}{|\nabla f|} =$ normal vector.

$\underline{x}' = \underline{x} - \alpha \hat{r} \underline{n} dt$ is the new position of a point at the $t+dt$, which is also on the surface.

- $f(\underline{x}', t+dt) = 0$ [new surface]

$$\rightarrow f(\underline{x} - \alpha \hat{r} \underline{n} dt, t+dt) = 0$$

Taylor series expansion about $\underline{x}$:

$$f(\underline{x}, t) + \frac{\partial f}{\partial t} dt - \nabla f \cdot \alpha \hat{r} \underline{n} dt + \Theta(dt^2) = 0$$

$$\frac{\partial f}{\partial t}(\underline{x}, t) = \alpha \hat{r} |\nabla f(\underline{x}, t)| \quad (50)$$

Shape of curve can be generated by $f(\underline{x}, t) = 0$